Experiment 2
Optoelectronic energy conversion

Table of contents

Page

1 Introduction 2
2 Basics 2
2.1 Radiometric quantities 2
2.2 Fundamentals of photovoltaics 3
2.3 Solar cells 9
2.4 Yield analysis 13
3 Experimental setup 15
3.1 Sun simulator 15
3.2 Spectrometer 16
3.3 Keithley Source/Measure Unit B2901A 16
4 Experiment with solar cells and a fan 17
5 Carrying out the experiment 17
6 Evaluation 18
7 Literature 18
8 Appendix 19

1 Introduction

The direct conversion of light energy into electrical energy is known as photovoltaics. Photons are absorbed in a so-called photovoltaic cell and generate free charge carriers, electrons and holes. These charge carriers are separated by a specific asymmetry in the electrical component and generate an electrical voltage at the external contacts via the resulting potential energy of the charge carriers. An externally connected load resistor leads to an optically - generated current and thus to an electrical power in this consumer. It is immediately apparent that the most important parameter of such a photovoltaic element is the power efficiency, i.e. the ratio of the generated electrical power to the incident optical power.

Most photovoltaic cells consist of semiconductors and absorb the radiation through the internal photoelectric effect. The separation occurs either through a heterojunction of different materials or for the most part through a pn-homo junction. Probably the best-known application of photovoltaics is the solar cell, which plays a central role in the discussion of renewable energies; it converts sunlight into electrical power.

In this experiment, the functioning of photovoltaic cells with a pn junction is investigated experimentally and theoretically. The characteristic curve of the cells used corresponds to that of a pn diode. Characteristic parameters of photovoltaics, such as the short-circuit current I_{SC} , the open-circuit voltage V_{OC} , the point of maximum available electrical power MPP, the fill factor FF and the power efficiency η, are determined from the characteristic curve. The loss mechanisms that reduce the power efficiency are of great importance. The discussion focuses on the one hand on the solar spectrum, which is important for solar cells, and on the other hand on various solar cell concepts that are currently being researched and developed. A radiation map is used to examine the economic efficiency of a solar system.

2 Basics

2.1 Radiometric quantities

When measuring electromagnetic radiation, a distinction is made between radiometry and photometry. Radiometry generally refers to the quantitative measurement of physical radiation quantities, while photometry is the measurement of visible light in units that are weighted according to the sensitivity of the eye. The physical quantities of radiometry ($SP(\lambda_0)$) therefore correspond to photometric quantities ($LT(\lambda_0)$), which are linked by the following relationship:

$$LT(\lambda_0) = K_m V(\lambda_0) \cdot SP(\lambda_0) \qquad (1)$$

$V(\lambda_0)$ is the relative eye sensitivity and K_m = 683.002 lm/W is the maximum luminous efficacy at v = 540 - 10^{12} Hz (λ = 555.016 nm). In photovoltaics, the radiometric quantities are used, the most important of which are listed in Table 2-1. For information, a comparison of the radiometric and photometric quantities is shown in the appendix.

Radiant energy	J; W-s	Q
Radiant power	W	$P = \frac{dQ}{dt}$
Radiant intensity	W-sr^{-1}	$P_\Omega = \frac{dP}{d\Omega}$
Irradiance (Intensity)	W-m^{-2}	$S = \frac{dP}{dA}$
Spectral irradiance	W-m^{-2}-nm^{-1}	$S_\lambda = \frac{d^2P}{dAd\lambda}$
Radiance	W-m^{-2}-sr^{-1}	$L = \frac{d^2P}{dA\cos(\theta)d\Omega} = \frac{dS}{dA\cos(\theta)}$

Tab. 2-1: Radiometric quantities

2.2 Basics of photovoltaics

2.2.1 pn diode

Photovoltaic cells are generally semiconductor components, more precisely pn or pin diodes, in which a photocurrent is generated by absorbing irradiated photons.
Due to their crystal structure, all semiconductors are characterized by the existence of an energy range that is forbidden for electron states. This energy gap W_G is the band gap and separates occupied from unoccupied states or the valence band from the conduction band, as the highest occupied state at T = 0K is the highest energy state below the forbidden range. Looking at the band model in momentum space, there are two possibilities for the absolute minimum of the conduction band and the absolute maximum of the valence band in relation to each other. If the minimum is at the same pulse as the maximum, the semiconductor is referred to as a direct semiconductor; if the minimum and maximum are at different pulses, the semiconductor is referred to as an indirect semiconductor. This is of crucial importance for the absorption and emission of light, as the conservation of momentum and energy must apply.

$$\text{Energie}: \quad \hbar\omega = W_A - W_E \geq W_G \tag{2}$$

$$\text{Impuls}: \quad \frac{\hbar\omega}{c} = \hbar\mathbf{k}_\mathbf{E} - \hbar\mathbf{k}_\mathbf{A} \tag{3}$$

Here ħω is the photon energy, ħω/c the photon momentum, ħk the electron momentum and the indices A and E stand for the initial and final states. As the photon momentum is very small, the following applies: $k_E \approx k_A$. This is why direct semiconductors absorb photons much better than indirect semiconductors. There is also a so-called absorption edge at $\hbar\omega = W_G$, which allows the band gap of a semiconductor to be determined by absorption.

Absorption reduces the intensity, which is dependent on the number of photons, of the incoming light as it passes through the crystal. The change in the intensity of the light dS is proportional to the so-called absorption coefficient α. The generally valid absorption law is obtained:

$$S_x = S_0 \cdot \exp(-\alpha x) \qquad (4)$$

This therefore means that the basic intensity S_0 decreases exponentially with the penetration depth. The absorption coefficient depends on the wavelength and determines the absorption length x_L , which corresponds to the value 1/α. As absorption produces both free states in the conduction band (electrons) and in the valence band (holes), these charge carriers must be spatially separated so that they contribute to the photocurrent.
For this reason, photovoltaic cells are equipped with a pn junction that separates the charge carriers by forming an internal electric field. The area in which the field is formed is called the space charge zone. The potential difference is the so-called diffusion voltage U_D . The course of the energy levels and the formation of the electric field are shown in Fig. 2-1.

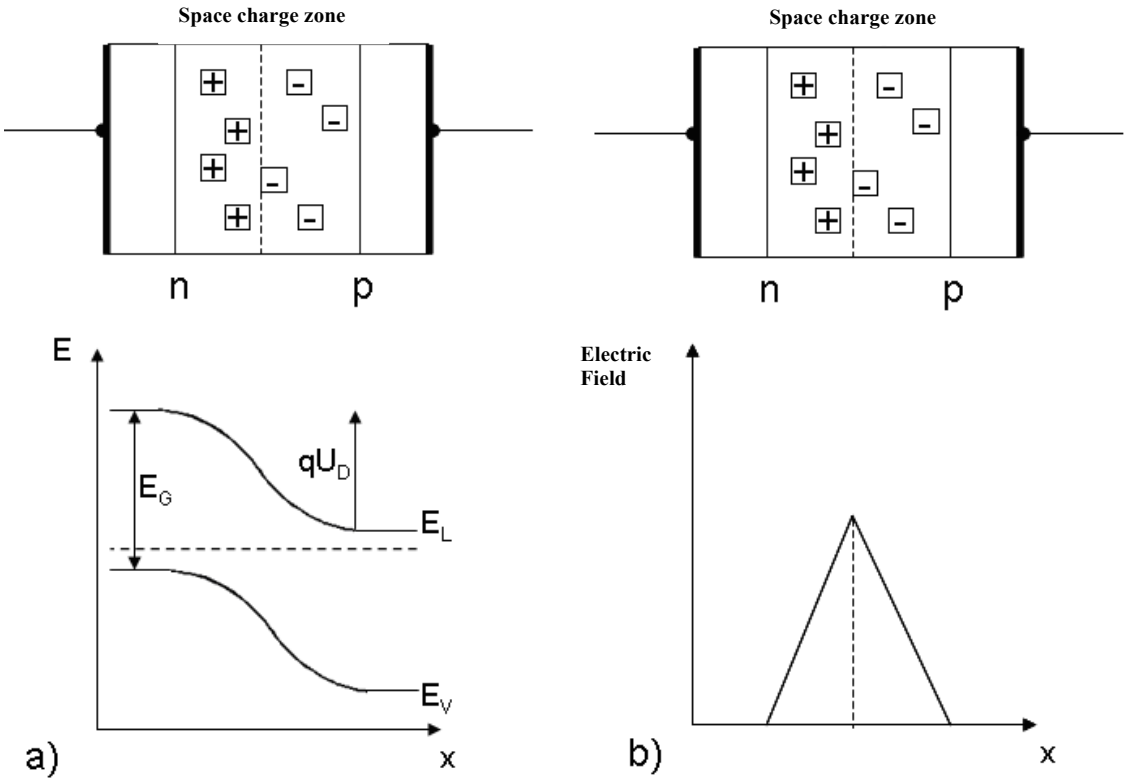


Fig. 2-1: Band diagram of a pn junction (a) and internal electric field (b)

2.2.2 Current-voltage characteristic

The I-V dark characteristic curve can be used to characterize diodes, for which the following relationship applies:

$$I_D = I_S \left(\exp\left(\frac{U}{n \cdot U_T} \right) - 1 \right) \qquad (5)$$

The value I_D indicates the total diode current that flows through the pn junction. This depends on the saturation blocking current I_S , the applied voltage U and the temperature or temperature voltage $U_T = k_B \cdot T/e \approx 26$ mV at 20° C. The value for the diode ideality factor n (also known as the emission coefficient) is usually between 1 and 2. The saturation blocking current depends on the temperature. The following applies:

$$I_S(T) \propto \exp\left(-\frac{W_g}{2 \cdot k_B \cdot T}\right) \qquad (6)$$

The band gap W_G can be determined by determining I_S for different temperatures and plotting it semi-logarithmically against the inverse temperature (in 1/K).

In the illuminated case, the total current I consists of a superposition of the photocurrent I_{SC} generated in the short circuit and the diode current I_D .

The following applies:

$$I = I_{SC} - I_D = I_{SC} - I_S\left(\exp\left(\frac{U}{n \cdot U_T}\right) - 1\right) \qquad (7)$$

If the total current is now plotted against the voltage, characteristic curves are obtained as shown in Fig. 2-2. This graph also shows some important solar cell parameters that are of central importance for characterization and are therefore explained in more detail below.

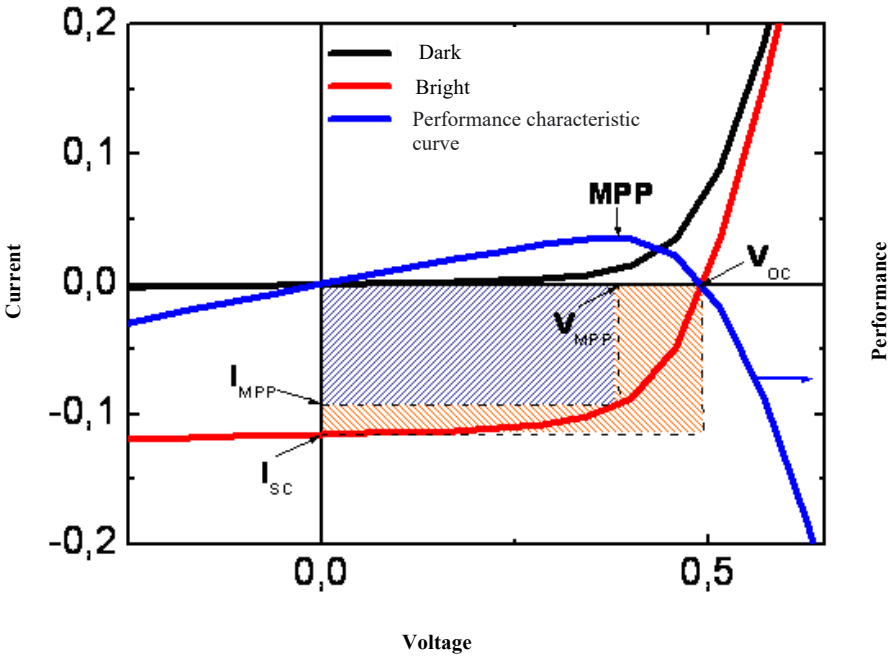


Fig. 2-2: Light/dark and power characteristics of a diode

Short-circuit current I_{SC}

The short-circuit current (= photocurrent) corresponds to the photocurrent generated in the short circuit. The load resistance is equal to 0 in this case. I_{SC} depends on the irradiated optical power and the external quantum efficiency η_{ext} :

$$I_{SC} = \eta_{ext} \cdot \frac{q}{h \cdot f} \cdot P_{opt} \qquad (8)$$

Open circuit voltage U_{OC}

The open-circuit voltage is the voltage applied to the diode when no current is flowing, i.e. when the load resistance is infinite. The following relationships result from (7):

$$I_{SC} = I_S \left(\exp\left(\frac{U_{OC}}{n \cdot U_T} \right) - 1 \right) \qquad (9)$$

and thus

$$U_{OC} = n \cdot U_T \cdot \ln\left(\frac{I_{SC}}{I_S} + 1 \right) \cong \frac{n \cdot k_B \cdot T}{q} \ln\left(\frac{I_{SC}}{I_S} \right) \qquad (10)$$

Maximum Power Point (MPP)

The maximum power point is the point at which the load resistance is optimized and the maximum electrical power (P_{MPP}) can be drawn from the cell. If the generated electrical power is plotted against the voltage, this is the maximum of the curve. For P_{MPP} this means

$$P_{MPP} = U_{MPP} \cdot I_{MPP} \qquad (11)$$

I_{MPP} is the current and U_{MPP} is the voltage at the maximum power point.

Fill factor FF

The fill factor is the ratio of the maximum extractable power P_{MPP} to the maximum possible electrical power P_{max} . In Figure 2-2, the shaded areas correspond to these two powers (blue: P_{MPP} ; red: P).$_{max}$

$$FF = \frac{P_{MPP}}{P_{max}} = \frac{U_{MPP} \cdot I_{MPP}}{U_{OC} \cdot I_{SC}} \qquad (12)$$

Power efficiency η

The power efficiency of the photovoltaic cell can now be determined from the previously determined variables by relating the electrical power to the optical power incident on the solar cell:

$$\eta = \frac{P_{MPP}}{P_{opt}} = \frac{FF \cdot U_{OC} \cdot I_{SC}}{P_{opt}} \qquad (13)$$

2.2.3 Loss mechanisms

On the one hand, there are losses in a solar cell due to the fact that not all photons that hit the surface of a photovoltaic cell contribute their full energy to power generation. On the other hand, there are electrical losses, e.g. relating to resistors inside the solar cell. The electrical losses are summarized in series resistance (R_S) and parallel resistance (R_P) (see 2.2.4, equivalent circuit diagram). These loss mechanisms therefore reduce the overall efficiency of a solar cell. The most important loss mechanisms are marked with a red arrow in Fig. 2-3 and are explained in more detail.

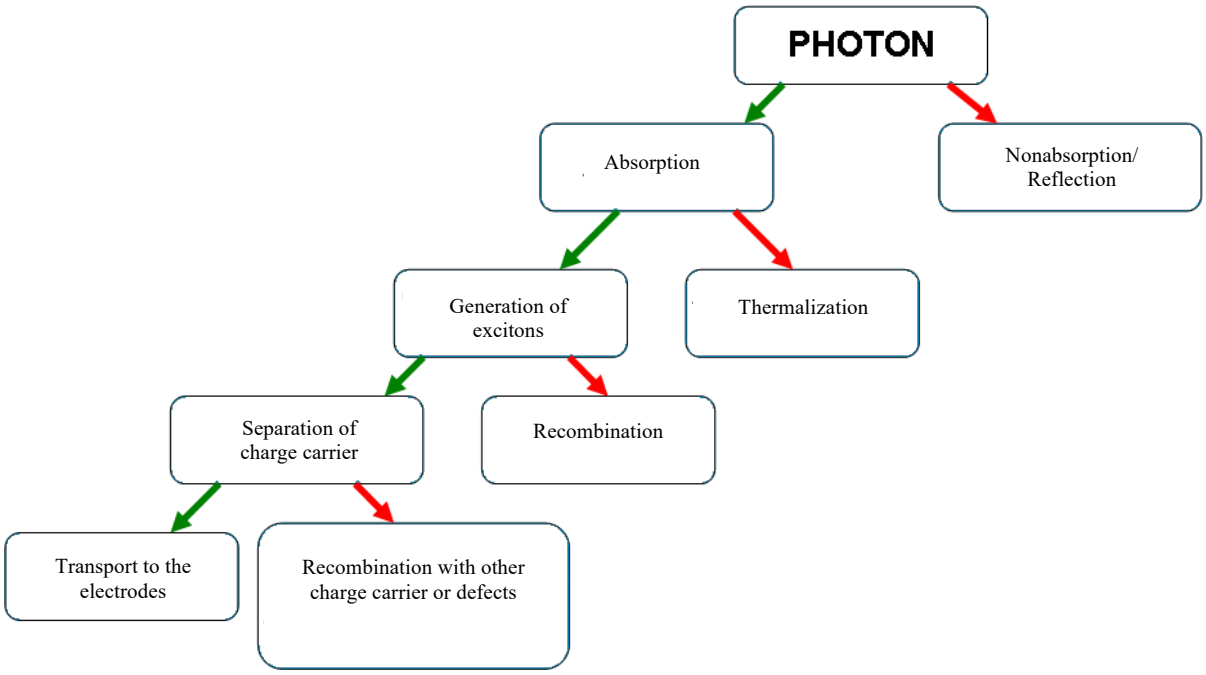


Fig. 2-3: Processes from photon to potential electrical power

Reflection:

Photons that hit the solar cell are reflected to a certain degree at the interface between the air and the solar cell. This degree of reflection depends on the refractive index difference between the air and the solar cell material as well as the so-called extinction coefficient κ. The losses due to reflection can be reduced by using an anti-reflective coating, for example.

Non-absorption:

Non-absorption is caused on the one hand by photons that cannot be absorbed in the semiconductor due to an energy $W < W_G$ and on the other hand by photons transmitted through

the semiconductor. The transmittance describes the proportion of photons that can penetrate the component.

Thermalization:

The thermalization of charge carriers is responsible for the existence of an optimum band gap in spectrally distributed light. Light with an energy greater than that of the band gap generates charge carriers within the bands of the semiconductor. These thermalize towards the band edge and release the excess energy (difference between the energy of the excited state and the band edge energy) as thermal energy to the crystal lattice.

Recombination:

Free charge carriers in the conduction band (electrons) recombine with free charge carriers in the valence band (holes) or via intermediate states (intrinsic or extrinsic interference sites). The energy released is either emitted as radiation (radiative recombination) or via other mechanisms as thermal energy (Auger recombination, Shockley-Read-Hall recombination).

Electrical losses:

The electrical losses of a solar cell are summarized in series and parallel resistance. The series resistance R_S is mainly made up of the ohmic resistance of the absorber material, the resistance between the absorber material and the metal contacts and the resistance of the metal contacts. Ideally, the following applies: $R_S \to 0$. The parallel resistance symbolizes the quality of the pn junction. For low losses, it should be as high as possible ( ). $R_P \to \infty$

2.2.4 Equivalent circuit diagram of a photovoltaic cell

The usual equivalent circuit diagram of a photovoltaic cell is shown in Fig. 2-4. In this case, the photovoltaic cell is replaced by a current source, diode, series resistor (R_S) and parallel resistor (R_P). The current source represents the generation of free charge carriers by light and the diode describes the pn junction in the photovoltaic cell. A characteristic curve in semi-logarithmic representation with designation of the individual parameters is shown in Fig. 2-5.

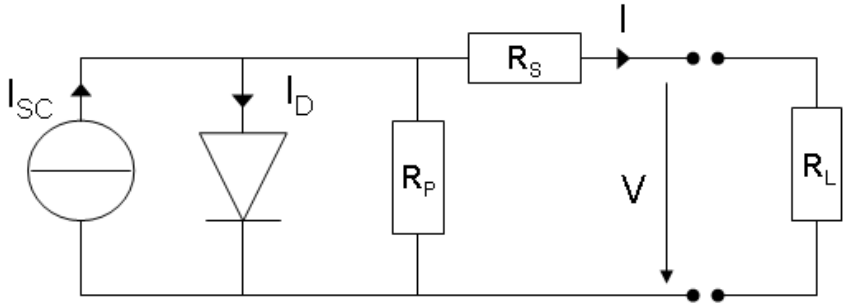


Fig. 2-4: Equivalent circuit diagram of a photovoltaic cell

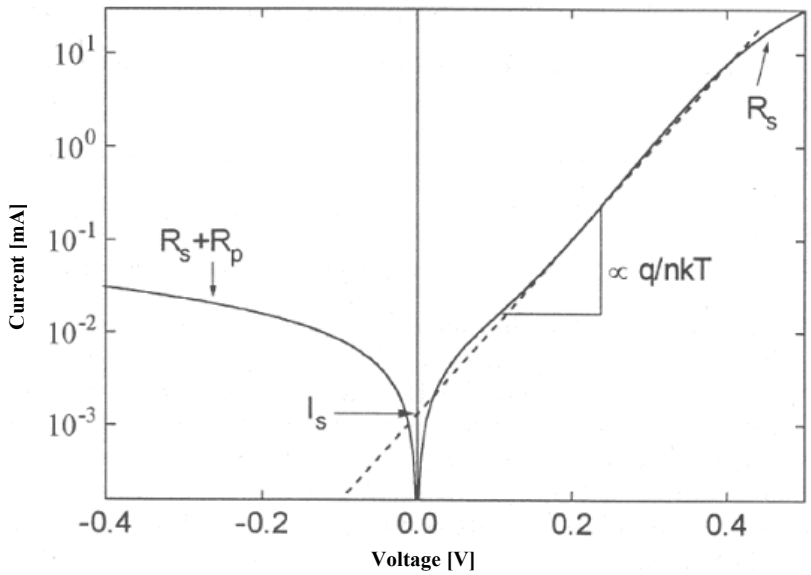


Fig. 2-5: IV dark characteristic curve (semi-logarithmic) of a photovoltaic cell

2.2.5 Dependence of the efficiency on the intensity

According to equation (13), the efficiency of a cell depends directly on the irradiated optical power. But how do the other parameters vary with the intensity? The spectral distribution of the light remains identical, so that the change in intensity can be regarded as a change by a fixed factor C. For equation (13), this means the substitution: $P_{opt} \rightarrow$ C - P_{opt} . To a first approximation, the fill factor is independent of the intensity ($FF \rightarrow FF$). The short-circuit current is equal in magnitude to the photocurrent resulting from the total generation of electron-hole pairs by absorption. At C-fold light intensity, the number of charge carriers generated changes to C-fold: $I_{SC} \rightarrow$ C - I_{SC} . If this is inserted into equation (10) for the open-circuit voltage, the result for U_{OC} is an increase by a factor of $\frac{\ln(C)}{\ln(I_{SC}/I_0)}+1$. Therefore, for $U_{OC} \rightarrow \left(\frac{\ln(C)}{\ln(I_{SC}/I_0)}+1\right)$ - U_{OC} . Substituting this into equation (13) gives the intensity dependence of the efficiency. After a conversion to change the power efficiency, the following applies:

$$\Delta\eta = \eta(\ln(C)) - \eta(\ln(1)) = const \cdot \ln(C) \qquad (14)$$

2.3 Solar cells

2.3.1 Solar spectrum

Solar cells convert light with a spectral distribution into electrical power. The task is therefore to optimize the cell to the distribution of the sun. The spectral distribution on the earth's surface

mainly depends on the path length that the light travels through the earth's atmosphere. The change in the spectrum is caused by the absorption of light in the atmosphere. The nomenclature AMx is commonly used for the light path (where AM = air mass or air mass passed through and x = x times the path through the earth's atmosphere). AM0 is therefore the light spectrum in space, which represents the absorption-free emission spectrum of the sun (see Fig. 2-6). It corresponds approximately to the emission spectrum according to Planck's radiation law of a black body with a temperature of T = 5900 K (surface temperature of the sun). The intensity is approximately S_0 = 1350 W/m², which is referred to as the solar constant.
AM1 is referred to as the spectrum that can be measured by the vertical incidence of light on the earth. The AM 1.5G spectrum (G = global radiation) is important for the characterization of solar cells, as it is defined as the standard for measurement in accordance with international standards and therefore provides direct comparability of solar cells worldwide. The total intensity for this spectrum is $S_{1.5G}$ = 1000 W/m².

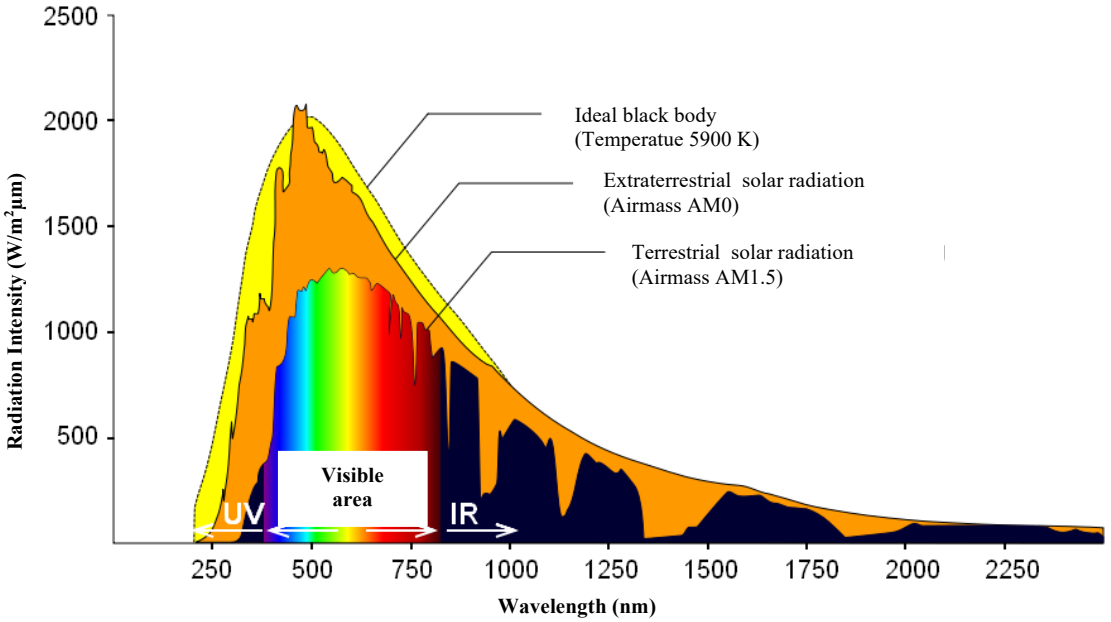


Fig. 2-6: Spectral distribution of solar radiation for different air masses compared to blackbody radiation at T = 5900 K

2.3.2 Solar cell concepts

Different absorbers in the solar cell are differently adapted to the solar spectrum, as can be seen in Fig. 2-7. Based on the absorber materials, solar cells are divided into different concepts, some of which are described below:

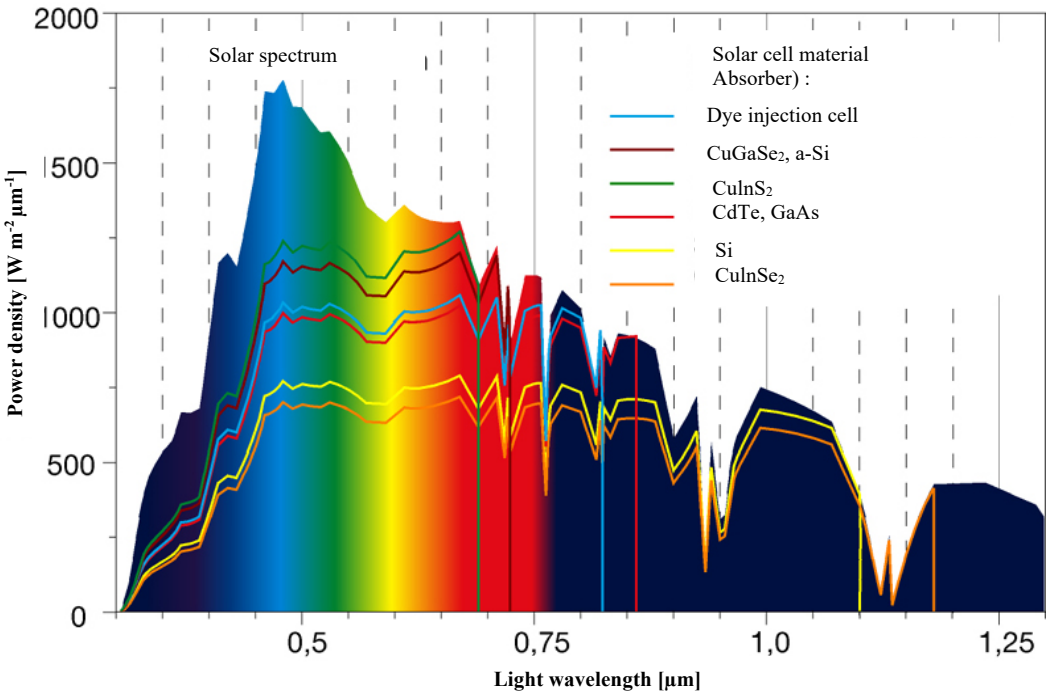


Fig. 2-7: Usable solar energy for different absorbers

c-Si cell (crystalline silicon)

The structure of a solar cell made of crystalline silicon is shown in Fig. 2-8. The blank is a p-doped silicon wafer, which is turned into a finished solar cell after various production steps. This is currently the most widely used type of solar cell. The advantages are high efficiency (compared to other concepts), a long service life with low degradation (reduction in efficiency over time) and a component without toxic substances. Weaknesses are the energy payback time (time in which the energy for production is generated by the cell itself) and the purchase price.

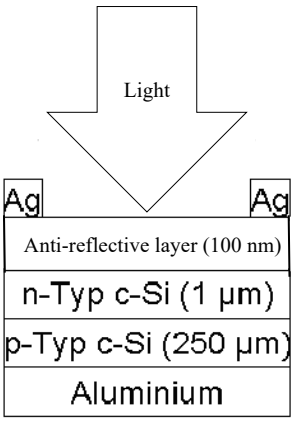


Fig. 2-8: Schematic structure of a c-Si solar cell

a-Si cell (amorphous silicon)

The structure of an amorphous cell (Fig. 2-9) shows on the one hand that it is a thin cell (≈1 µm; without glass substrate) compared to a c-Si cell (≈300 µm) and on the other hand that the basic structure is a pin junction. Due to the material savings and cheaper production, there are advantages in the energy payback time and the purchase price. However, the efficiency is lower than that of c-Si cells and the service life and degradation are also worse.

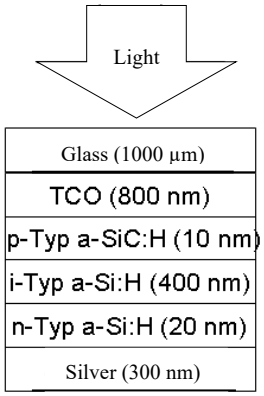


Fig. 2-9: Schematic structure of an a-Si solar cell

CIS cell (copper indium gallium diselenide)

The total thickness of the active layers ($Cu(In,Ga)Se_2$, CdS, Mo, ZnO) of this type of solar cell is also in the range of a few micrometers and therefore the cells are cheaper in terms of purchase price and energy payback time than c-Si cells. In this case, the pn junction is a heterojunction (different materials). However, the heavy metal cadmium (Cd) is used and the efficiency does not reach that of c-Si cells (Fig. 2-10).

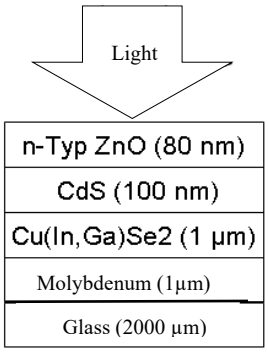


Fig. 2-10: Schematic structure of a CIS solar cell

After this brief description of different types of solar cells, it has become clear that all concepts have advantages, but also disadvantages compared to other concepts. Further research is therefore needed to develop the perfect solar cell.

2.4 Yield analysis

The annual energy yield of a solar cell depends on the cumulative irradiated light output and the efficiency η. The cumulative output depends on the duration of the irradiation. This means that the annual energy yield EE_{Jahr} can be determined as the integral of the product of the irradiated optical power of the sun (P_{Sonne}) and the cell efficiency over the duration of a year:

$$EE_{Jahr} = \int_{Jahr} P_{Sonne}(t) \cdot \eta\left(P_{Sonne}(t)\right) dt \qquad (15)$$

Since the irradiance of the sun and thus the irradiated optical power vary continuously over time, averaging over different time periods, for example, makes sense. The appendix shows the cumulative power (P) for the winter and summer half-years for North Rhine-Westphalia. The energy yield for the year, e.g. for Duisburg (DU), can then be calculated from this for an assumed constant irradiance of 1000 W/m² (AM 1.5G):

$$EE_{S_{1000W/m^2}} = \sum_{i=1}^{2} P_i \cdot \eta\left(S_{1000W/m^2}\right) \qquad (16)$$

If one now calculates with different power efficiencies at X-fold different irradiance levels and a distribution of irradiance according to Table 2-2, then the annual energy yield can be calculated from

$$EE_{Jahr} = \sum_{n=1}^{X} \sum_{i=1}^{2} P_i \cdot a_n\left(S_X\right) \cdot \eta\left(S_X\right). \qquad (17)$$

Irradiance S_X	Share of cumulative output a_S in the winter half-year in %	Share of cumulative output in the summer half-year in %
up to 50 W/m²	28,5	16,5
up to 100 W/m²	27,4	13,2
up to 200 W/m²	24,0	18,6
up to 300 W/m²	11,5	13,1
up to 500 W/m²	7,2	18,8
up to 1000 W/m²	1,4	19,8

Tab. 2-2: Distribution of irradiance levels for the winter and summer half-year in Duisburg

For comparison, the following values are given for Tarifa (Spain), almost the southernmost point on the European continent. P_{Jahr} is 1919.22 kWh/m² and is divided into 654.55 kWh/m² in the winter half-year and 1264.67 kWh/m² in the summer half-year. The irradiance is distributed as follows:

Irradiance S_X	Share of cumulative output in the winter half-year in %	Share of cumulative output

		in the summer half-year in %
up to 50 W/m²	11,4	8,9
up to 100 W/m²	9,2	6,6
up to 200 W/m²	14,7	9,8
up to 300 W/m²	12,3	8,7
up to 500 W/m²	22,0	15,0
up to 1000 W/m²	30,4	51,0

Tab. 2-3: Distribution of irradiance levels for the winter and summer months in Tarifa

To enable a direct comparison for the purchase of solar cells, the electrical power generated by the corresponding commercial solar cells is given in W_p (Watt Peak). This specification refers to the output of the solar cell under the following standardized test conditions (STC, Standard Test Condition):

- Spectral intensity distribution according to AM 1.5G
- Irradiance: 1000W/m²
- Cell temperature: 25°C (±2°C)

3 Experimental setup

This chapter describes the most important measuring instruments used in this experiment and how to use them. These include the solar simulator for simulating the AM 1.5G sunlight spectrum, the spectrometer for spectral measurement of the solar simulator light and the Keithley Sourcemeter for measuring the IU characteristics.

3.1 Sun simulator

A sun simulator ensures constant, defined lighting conditions in the measuring plane of the solar cell. This makes it easy to compare different solar cells and their measurement results. The simulator used here consists of an LED array with 960 LEDs:

Dimensions: 22 cm x 24 cm
30 x 32 = 960 LEDs

LED types: 480 x Nichia NSPL500DS warm white
480 x Nichia NSW500DS cold white

Supply voltage: 24 Volt DC
Optical output power: max. 6.25 watts
Optical irradiance: max. 300 W/m²
Electrical power loss: max. 76.8 watts
Intensity setting via pulse width modulator (PWM)

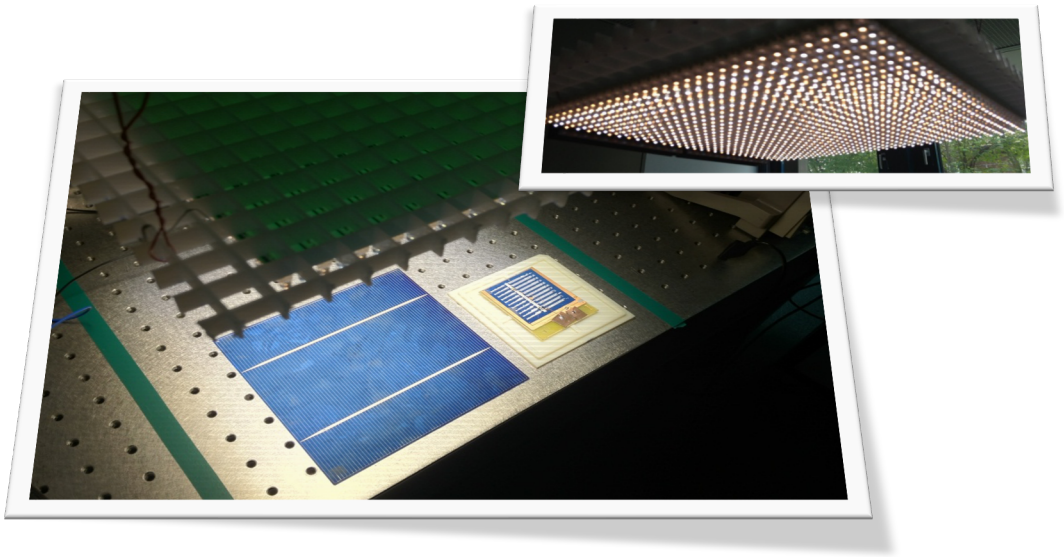


Fig. 3-6: LED array of the solar simulator

3.2 Spectrometer

The getSpec-3648 spectrometer from Sentronic (see Fig. 3-7) is used to record the spectral distribution of the light emitted by the sunlight simulator. The light is coupled into a glass fiber via a diffuser and guided through a grating to a CCD line detector. It measures in the wavelength range from 360 nm to 1100 nm and has a resolution of 2.4 nm. The intensity is displayed in the unit $\frac{\mu W}{cm^2\ nm}$ unit.



Fig. 3-7: Spectrometer getSpec-3648

3.3 Keithley Source/Measure Unit B2901A

To determine the electrical properties of the photovoltaic cell, a combination measuring instrument is used which serves as a voltage source and can simultaneously measure the electrical current. A graphical evaluation of the current-voltage characteristic curve under different lighting conditions is also possible.



Fig. 3-8: Keysight Source/Measure Unit B2912A

4 Interconnection of solar cells with a fan

In this section, the bypass characteristic of a reverse-biased diode is investigated. In order to visualize the currents that occur, an LED is used as a protective diode. For the setup, a solar cell and a light-emitting diode are first connected in parallel, whereby the polarity of the components must be observed. The fan is then connected in series with this circuit. In the next step, a second parallel circuit consisting of solar cell and LED is added, resulting in the structure shown in Figure 3-9.

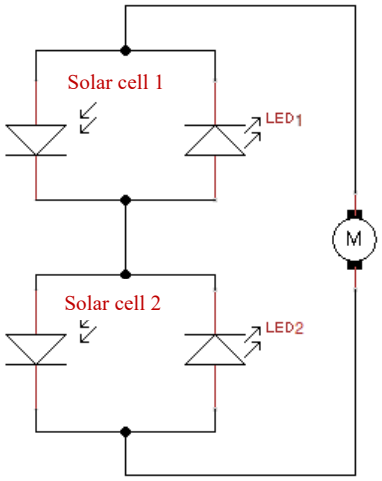


Fig. 3-9: Wiring of the solar cells with protective diodes and fan

5 Test execution

(a) Use the spectrometer to measure the spectral irradiance of the Solar simulator in µW/(cm² - nm)
(b) Determine the area of the solar cell to be measured
(c) Record IU characteristics (dark) at different temperatures
(d) Record IU characteristics (bright) at different intensities
(e) Set up the circuits described in section 4 and investigate the effects of a change in intensity (25%, 50%, 75% and 100%) on the speed of rotation of the fan and the brightness of the LEDs (bypass diodes)

Note: Experiments (c) and (d) must be planned (with the help of the supervisor) in such a way that reliable statements can be made about the band gap of the semiconductor material used and the intensity dependence of the power efficiency.

6 Evaluation

I. Compare the measured spectral irradiance with the solar spectrum AM 1.5G

II. Integrate the measured spectral irradiance and compare the value with the value of the measured irradiance and the standardized irradiance of the solar spectrum AM 1.5G

III. Determine the following diode parameters using the IU characteristic curves (dark):
   a. Blocking saturation current I_S
   b. Diode ideality factor n
   c. Series/parallel resistance R_S and R_P
   d. Band gap W_G

IV. Determine the following solar cell parameters using the IU characteristic curves (light):
   a. Short-circuit current I_{SC}
   b. Open circuit voltage U_{OC}
   c. Current in Maximum Power Point I_{MPP}
   d. Voltage at maximum power point U_{MPP}
   e. Maximum extractable power P_{MPP}
   f. Incident radiant power P_{opt}
   g. Fill factor FF
   h. Efficiency η

V. Apply η against the intensity and discuss the progression.

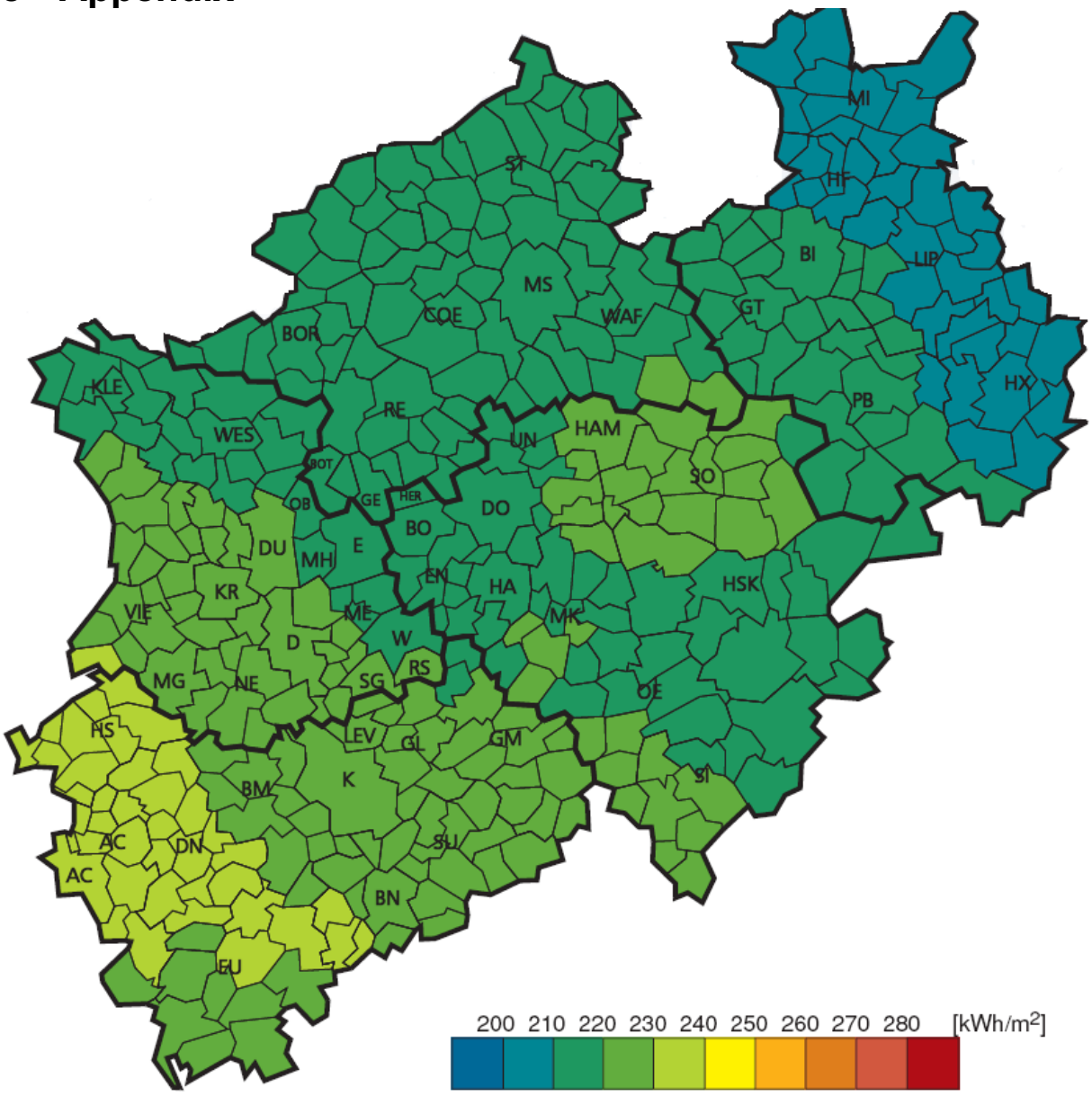
VI. Summarize the results from section 4 (interconnection of solar cells with fan) in tabular form and discuss the visual impressions at different illuminance levels.

VII. Determine the annual energy yield and the cost of a kWh generated with a service life of 20 years and acquisition costs of 4000 EUR/kWp (including all expected maintenance costs) of the measured solar cell
   a. by means of a radiation map
   b. using a radiation map and Duisburg radiation distribution
   c. using the specified parameters from Tarifa (Spain)

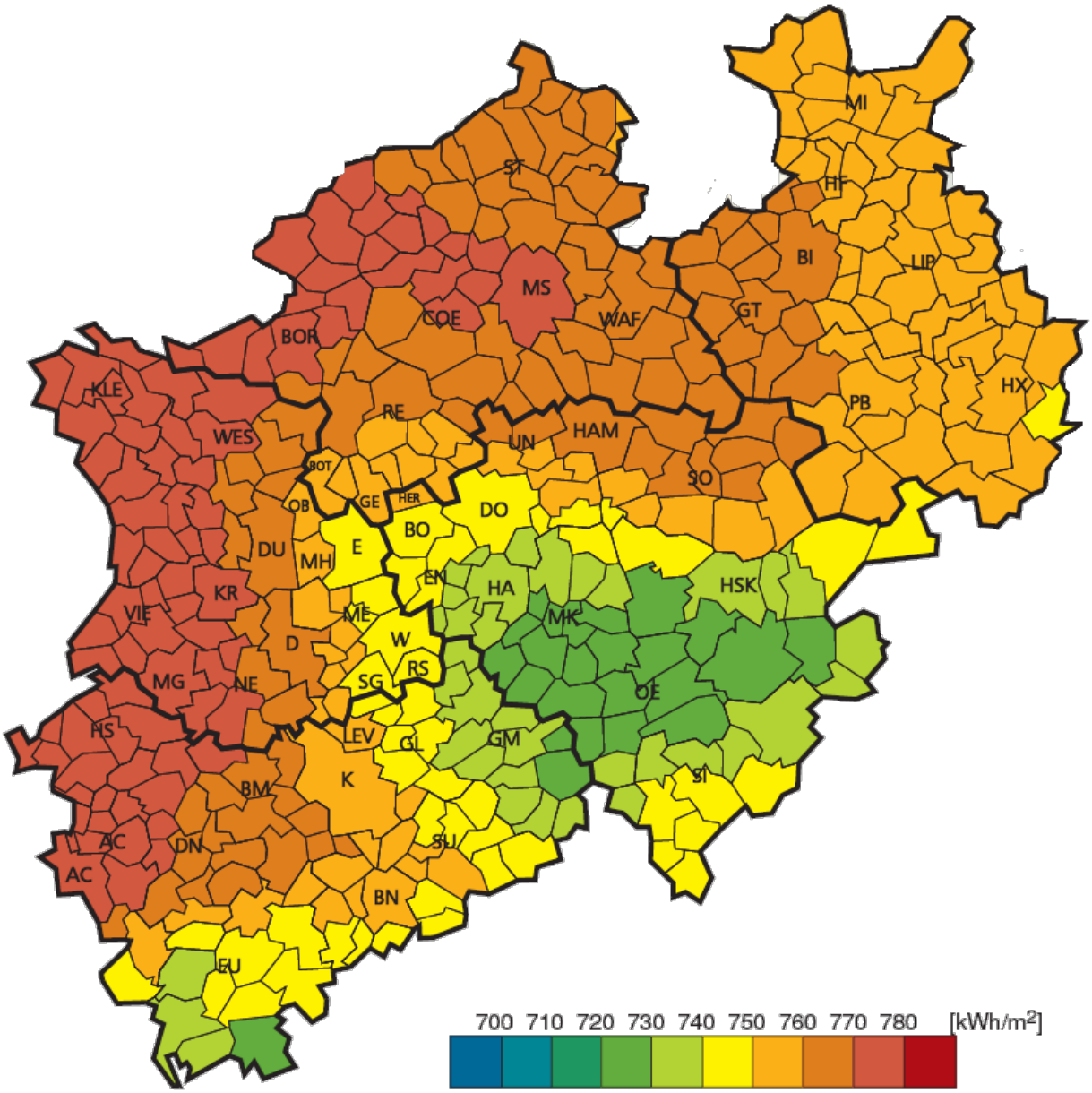
7 Literature

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8 Appendix



Solar radiation map winter half-year (October to March)



Solar radiation map summer half-year (April to September)

Radiometric quantities			Lighting parameters	
Designation	Unit	Formula symbol	Unit	Designation
Radiant energy	J; W·s	Q	lm·s	Amount of light
Radiant power	W	$\Phi = \frac{dQ}{dt}$	lm	Luminous flux
Beam strength	$W \cdot sr^{-1}$	$I = \frac{d\Phi}{d\Omega}$	cd; $lm \cdot sr^{-1}$	Luminous intensity
Irradiance (intensity)	$W \cdot m^{-2}$	$E = \frac{d\Phi}{dA}$	lx; $lm \cdot m^{-2}$	Illuminance
Spectral irradiance	$W \cdot m^{-2} \cdot nm^{-1}$	$E_\lambda = \frac{d^2\Phi}{dA\, d\lambda}$		
Radiance	$W \cdot m^{-2} \cdot sr^{-1}$	$L = \frac{d^2\Phi}{dA \cos\theta\, d\Omega} = \frac{dI}{dA \cos\theta}$	$cd \cdot m^{-2}$; $lm \cdot m^{-2} \cdot sr^{-1}$	Luminance

Comparison of radiometric and photometric quantities